

25 B $F = BIL$

Gradient of force-current graph = $BL = \frac{10 \times 10^{-3}}{5} = 2.0 \times 10^{-3}$

$\varepsilon = BLv = BL(\omega x_0) = 2.0 \times 10^{-3} (630)(1.5 \times 10^{-3}) = 1.86 \times 10^{-3} \text{ V}$

26 D $D = \frac{v^2}{A^2} = \frac{1}{A^2} \left(\frac{d\Phi}{dt} \right)^2 = \frac{1}{A^2} \left(\frac{d}{dt} BA \right)^2 = \frac{B^2}{A^2} \left(\frac{dA}{dt} \right)^2$ where $\frac{dB}{dt}$ = gradient of B vs graph